

tomisable DSP accelerators, logic blocks to support many market segments or customers with a single chip, reconfigurable base-station digital front-ends, programmable parsers, reconfigurable protocols and more. EFLX can provide a ton of I/Os—632 per direction per tile in a 7 x 7 array that adds up to big numbers even after tile-to-tile and memory interconnects. In the MCU segment, EFLX blocks can be configured as something that amounts to a smart crossbar.

EFLX, a fully-functional FPGA

EFLX FPGA has parallel data paths running vertically and interfacing to peripherals laterally such as the advanced extensible interface AXI Slave, which connects to ARM advanced MCU bus architecture. Basic part is an FPGA; interconnect and I/O block is called EFLX-2.5K when built on TSMC 28HPM. The 28nm high-performance mobile (HPM) computing provides high performance for mobile applications to address the need of applications requiring high speed. Such technologies can provide the highest speed among 28nm technologies. It gives implementer 2520 LUTs, 5040 flip-flops, eight clocks, 632 inputs and 632 outputs.

The second variant is called EFLX DSP core, and its main addition is a DSP block.

EFLX blocks can be used anywhere in a chip for connection to RAM and I/O, in the data path or on a processor bus. It can integrate any kind of RAM of any width, size, having single or dual port, ECC, parity or neither. Memory can be put between cores and controlled with unused I/O.

EFLX memory compiler maps RTL code onto the array including RAM. EFLX blocks are programmed with RTL. Both architectures can be tiled together in up to a 7 x 7 array, and, if needed, provide GDS, LIB, LEF and Verilog files. The

incremental die cost is less than five cents per k-input-look-up tables (k-LUTs), which is much less than packaged FPGAs.

If memory is needed, interconnects between the tiles can have DRAM blocks added to these. Instead of wiring up I/Os from one tile to the next directly, memory is placed between tiles, and I/Os on the periphery of the tile group take over I/O duties for the entire set.

28nm is the first generation that foundry industry starts to use high-k metal gate (HKMG) process, which represents TSMC. Still, poly/oxnitride process is offered to meet customers' time-to-market need. It is ideal for low standby power applica-

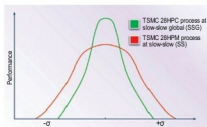


Fig. 1: SSG corner signoff provides 10 to 15 per cent performance boost over 28HPM SS corner signoff

tions such as mobile baseband. In a 28nm process, fast control logic can run at 525MHz or more, and pipelined finite impulse response filters (FIRs) can run up to 450MS/s or faster.

In 40nm, multiple speed/power tradeoffs are supported to address almost all possible requirements. The 28nm low-power with high-k metal gates (HPL) technology adopts the same gate stack as HP technology while meeting more stringent low-leakage requirements with a trade of performance speed.

The 28nm family also provides a wider variety of metal options to support a broad range of product applications for better trade-off between performance and density. It only requires six metal layers instead of the traditional ten or more in conventional mesh FPGAs, and

offers 1.8 times higher logic density, thanks to optimised interconnect.

TSMC offers three processes at 28nm. These are 28LP, 28HP and 28HPL processes. Millions of production wafers have come out of TSMC's 28nm processes. In TSMC, 28HPM/C fast control logic can run at 525MHz or more, and pipelined FIRs can run at 450MS/s or faster.

Designers can improve SoC performance by using global slow and fast (SSG, FFG) signoff corners. These are enabled by TSMC's tighter process controls with 28HPC/HPC plus over 28LP/HP/HPL/HPM. Improved performance enables the use of lower-drive (smaller) logic cells to close critical-timing paths.

28LP process. 28LP process has 20 per cent speed improved over 40LP process at the same leakage/gate.

28HP process. 28HP process targets CPU, GPU, FPGA, PC, networking and consumer electronics applications. It supports 45 per cent speed improvement over 40G process at the same leakage/gate. With higher performance coverage, 28HPM is ideal for many applications in networking and high-end smartphone/mobile consumer products.

28HPL process and 28HPM.

28HPL process reduces both standby and operation power by more than 40 per cent. Compared with TSMC's 28LP, 28HPC provides ten per cent smaller die size and more than 30 per cent power reduction at all levels of speed. A comprehensive 28HPC IP ecosystem is also built and compatible with 28HPM, accelerating time-to-market for customers. 28HPC is also ideal for many applications in mid- and mid-to-low end smartphones, tablets and mobile consumer products.

EFLX cores

EFLX core is memory type and needs to be TSMC 28HPC-compatible. TSMC provides high-performance compact mobile computing, the most energy-efficient and high-perfor-